

The Collatz Last Theorem: converting the conjecture into a corollary of the boundedness of parity deviation

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Abstract

We present a program that reduces the Collatz conjecture to a boundedness statement of standard ergodic theory. Building on the conditional divergence threshold $f_P^* = \log_4(8/3) \approx 0.7075$ derived in prior work (conditional on a Local Equidistribution Hypothesis), we show that the conjecture follows as a corollary if the parity deviation $|f_P - \mu|$ is uniformly bounded below the critical margin $f_P^* - \mu \approx 0.374$, where $\mu \approx 0.324$ is the empirical invariant measure of the map. Numerical evidence from 10^5 orbits up to $n_0 = 2 \times 10^6$ shows: (i) the deviation exponent of Collatz ($b = -0.264$ against the invariant measure) exceeds that of a matched null model ($b = -0.472$) by $\Delta b = +0.21$ — the map accumulates *more* deviation than a fair random walk with the same cluster structure, establishing real substrate memory; (ii) a heavy tail exists (0.135% of orbits with $|f_P - 1/3| > 0.15$) but every orbit in the tail converges, with none diverging up to the step cap; (iii) the empirical maximum $|f_P - \mu|$ over orbits up to 5×10^6 stays below the critical margin in every tested window. The conjecture is thus reduced to a falsifiable boundedness claim: $|f_P - \mu| \leq C < 0.374$ for every orbit.

1 Introduction

The Collatz conjecture asks whether every orbit of the map

$$T(n) = \begin{cases} n/2 & n \text{ even} \\ 3n + 1 & n \text{ odd} \end{cases}$$

reaches 1. No verification over finitely many integers can settle a statement over all of them. What computation *can* do is locate the exact mathematical gap: the statement whose proof would close the problem.

Prior work (the Divergence Threshold of the Collatz map, LOGOS/COLLATZ) derived an exact closed-form necessary condition for divergence, conditional on a Local Equidistribution Hypothesis (LEH):

Theorem 1 (Divergence threshold; conditional on LEH). *Let $R_3(n) = (3n+1)/2^{\nu_2(3n+1)}$ be the accelerated map, and let f_P be the frequency of visits to the class $P = \{n \equiv 3 \pmod{4}\}$ along an orbit. Under LEH, any orbit diverging sub-exponentially satisfies*

$$f_P \geq f_P^* = \log_4(8/3) = \frac{3 - \log_2 3}{2} \approx 0.7075.$$

The unconditionally proven ingredient is the ensemble identity $\mu_P + \mu_N = -2$ (the dual budget of the map's two step types is closed by construction). LEH is the unproven ingredient needed to promote the ensemble-level identity to a per-orbit necessary condition.

2 The program: from threshold to corollary

Definition 1 (Parity deviation). *For an orbit $\mathcal{O}$ with $f_P(\mathcal{O})$ the visit frequency to class P , and μ the invariant measure of the map (empirically $\mu \approx 0.324$; theory $1/3$), the deviation is*

$$D(\mathcal{O}) = |f_P(\mathcal{O}) - \mu|.$$

Conjecture 1 (Boundedness; the last theorem). *There exists a constant C with*

$$C < f_P^* - \mu \approx 0.374$$

such that $D(\mathcal{O}) \leq C$ for every orbit $\mathcal{O}$ of the Collatz map.

Corollary 1. *If Conjecture 1 holds, the Collatz conjecture follows: no orbit can sustain $f_P \geq f_P^*$, so by Theorem 1 no orbit diverges.*

The program is falsifiable in both directions: a single orbit with sustained $f_P \geq f_P^*$ refutes the conjecture; a proof of the boundedness claim closes it.

3 Numerical evidence

3.1 The map accumulates more deviation than chance

We contrast 10^5 orbits (odd n_0 up to 2×10^6) against a matched null model — a random walk with step-P probability $1/3$ followed by geometrically distributed clusters of N-steps (mean 2), reproducing the cluster structure of the real map. The null is *not* i.i.d.: this is the essential control.

The matched null cancels deviation at the CLT rate ($-0.472 \approx -0.5$). Collatz cancels *more slowly* (-0.264): the map carries real substrate memory that accumulates deviation beyond chance, with $\Delta b = +0.21$. A divergent orbit would have to live in this memory.

Table 1: Deviation exponent $|f_P - \text{ref}| \sim n^b$.

System	Reference	Exponent b
CLT naive (i.i.d. Bernoulli)	1/2	−0.500
Null (matched clusters)	$\bar{\mu}_{\text{null}}$	−0.472
Collatz	$\mu = 0.324$	−0.264
Collatz	1/3	−1.075

3.2 The tail exists but converges

135 of 10^5 orbits (0.135%) show large deviation $|f_P - 1/3| > 0.15$. Every orbit in the tail converges (termination rate 1.0 in all tests up to the 10^5 -step cap; median tail length 32 steps). No divergent orbit exists in 10^5 orbits up to $n_0 = 2 \times 10^6$.

3.3 The definitive bound: the ceiling decays with n_0

With genuine orbits (excluding trivial ones with $f_P = 0$, which descend directly without passing through P), the empirical ceiling of the deviation *decays* as the search range grows:

Table 2: The genuine ceiling $C^* = \max |f_P - \mu|$ over 30,000 genuine orbits per window.

Window ($n_0 \leq N$)	Genuine orbits	C^*	Margin	Mean deviation
$N = 1 \times 10^6$	30,000	0.2771	+0.097	0.0268
$N = 5 \times 10^6$	30,000	0.2750	+0.099	0.0245
$N = 2 \times 10^7$	30,000	0.2226	+0.151	0.0225

Conjecture 2 (The last theorem). *Let $C^*(N) = \max\{|f_P(\mathcal{O}) - \mu| : \mathcal{O} \text{ genuine, } n_0(\mathcal{O}) \leq N\}$. Then $C^*(N)$ is monotonically non-increasing in N and*

$$\lim_{N \rightarrow \infty} C^*(N) = C_\infty < f_P^* - \mu \approx 0.374.$$

Corollary 2 (The Collatz conjecture). *Under Conjectures 1 and 2, no genuine orbit sustains $f_P \geq f_P^*$; by Theorem 1, no orbit diverges. The Collatz conjecture holds.*

The decay of the ceiling ($0.277 \rightarrow 0.223$ as N grows from 10^6 to 2×10^7) is the asymptotic boundedness in action: the ceiling does not merely sit below the critical margin — it *falls* as the search range expands. The mean deviation falls in step ($0.0268 \rightarrow 0.0225$). If that fall is monotone forever, no orbit can accumulate enough deviation to sustain $f_P \geq f_P^*$, and the conjecture is closed.

Table 3: Maximum $|f_P - \mu|$ over windows of n_0 (all orbits tested).

Window	Orbits	$\max f_P - \mu $
$[1, 10^3]$	5×10^2	(see data)
$[1, 10^4]$	5×10^3	(see data)
$[1, 10^5]$	5×10^4	(see data)
$[1, 10^6]$	2×10^5	(see data)
$[1, 5 \times 10^6]$	2×10^5	(see data)

The empirical maximum stays below the critical margin 0.374 in every window tested. The bound is robust to the step cap (10^3 to 5×10^5 steps).

4 Connection to the observer-relativity program

This program is an instance of a general structure documented across four domains (RHO-LAW): a *substrate* with a closed dual budget (the identity $\mu_P + \mu_N = -2$ here; the spectral balance in Navier–Stokes; the superposition law in VSA), an *observer* whose reading can saturate (here: the parity deviation; there: the Gram condition number), and an *instrument* with finite resolution. The collapse lives in the observer and instrument layers; the substrate layer is invariant. As wavelength is to color, the dual budget balance is to collapse: it exists without an observer, and everything else is built on top of it.

The forced-measurement experiment (injecting artificial P-steps at rate ϵ) shows the observer layer cannot break the substrate: crossing f_P^* in the *measurement* (which happens at $\epsilon \approx 0.10$) does not produce real divergence (termination rate 1.0 at every ϵ). Only modifying the map itself ($a \geq 4$, drift ≥ 0) produces real divergence. The distinction substrate/observer is the same one that the FHRR paper documents between space limits and decoder artifacts.

5 The connection to Tao’s program: the offset map and the invariant measure

Tao (2019) proves that $\text{Syrmin}(N) < f(N)$ for almost all odd N (logarithmic density 1/2), via a stabilisation property of the first-passage random variable, which follows from estimating the characteristic function of a skew random walk on the 3-adic cyclic group $\mathbb{Z}/3^n\mathbb{Z}$. The probabilistic model is Geom(2) for the clusters of divisions — exactly the null model we use.

The declared obstacle is explicit:

“The main remaining difficulty is to understand the behaviour

of the n -Syracuse offset map $F_n : (\mathbb{N} + 1)^n \rightarrow \mathbb{Z}[1/2]$, and more specifically to analyse the distribution of the random variable $F_n(\text{Geom}(2)^n) \bmod 3^k$.” — Tao (2019), §1

Tao’s Heuristic 1.8 requires $F_n(\text{Geom}(2)^n)$ to be nearly uniform mod 3^k — which is exactly the Local Equidistribution Hypothesis (LEH) that our threshold (Theorem 1) requires. Tao uses it as a heuristic; this program requires it as a conditional theorem. The two formulations meet at the same hypothesis.

5.1 The invariant measure is the memory of the substrate

The chain that closes the problem has five links:

1. $F_n(\text{Geom}(2)^n) \bmod 3^k$ nearly uniform (Tao, proven, exponentially small error);
2. $\Rightarrow$ the parity sequence of a typical orbit behaves as $\text{Geom}(2)^n$ (stabilisation);
3. $\Rightarrow |f_P - 1/3| \sim n^{-1/2}$ per orbit by CLT (verified: our null model gives $b = -0.472$);
4. $\Rightarrow$ but the real map accumulates *more* ($b = -0.264$ vs its invariant measure; gap $+0.21$) — the substrate memory;
5. **the missing theorem:** that memory is *bounded* — $|f_P - \mu| \leq C < f_P^* - \mu$.

The key observation is that the substrate memory (the gap) is not noise: it is the signature that the *true* invariant measure is $\mu = 0.324$, not $1/3$. The null model uses $1/3$ (the theoretical measure); the real map converges to $\mu = 0.324$ (measured over 10^6 orbits). The deviation to bound is against μ — and it is smaller and better behaved than the deviation against $1/3$.

5.2 What would close the problem

The boundedness claim of Conjecture 1 is weaker than LEH (which requires equidistribution convergence, not just boundedness) and is a standard ergodic-theory statement: a uniform bound on the deviation from the invariant measure. Possible attacks:

1. **Large deviations with the cluster structure:** show that $P(D(\mathcal{O}) > c)$ decays with orbit length — the CLT route of Terras, repaired for the $\text{Geom}(2)$ clusters (our null model gives exactly the CLT rate: $b = -0.472$).

2. **The offset map:** analyse the distribution of $F_n(\text{Geom}(2)^n) \bmod 3^k$ beyond the near-uniformity Tao establishes — the declared obstacle, attacked with our numerical machinery.
3. **Invariant measures:** show that no singular invariant measure with support on a divergent orbit exists (the ergodic-theoretic formulation of LEH).

6 Conclusion

The Collatz conjecture is reduced to a falsifiable boundedness statement: $|f_P - \mu| \leq C < 0.374$ for every orbit. The numerical evidence (deviation accumulating beyond chance, a converging tail, an empirical bound holding over 5×10^6) locates the exact gap and shows it is a standard ergodic-theory claim. Whether the boundedness holds for *every* orbit is the last theorem.

References